

Student Rank & Percentile — Line-by-Line Calculation

1. Given Data

Student	Score	Point Score
Student_1	56	8
Student_2	72	14
Student_3	68	4
Student_4	91	11
Student_5	85	5
Student_6	77	9
Student_7	63	15
Student_8	95	6
Student_9	81	10
Student_10	74	2
Student_11	88	12
Student_12	69	3
Student_13	59	7
Student_14	93	13
Student_15	80	1

2. Total Number of Students

There are 15 students.

N = 15

3. Rank Formula

Rank is calculated from the Score, with the highest Score receiving Rank 1.

Excel formula: =RANK.EQ(B2,\$B\$2:\$B\$16,0)

The final argument 0 means descending order: highest score = Rank 1.

4. Ranking — Line by Line

Step	Student	Score	Rank calculation	Rank
1	Student_8	95	Highest score	1
2	Student_14	93	2nd highest score	2
3	Student_4	91	3rd highest score	3
4	Student_11	88	4th highest score	4
5	Student_5	85	5th highest score	5
6	Student_9	81	6th highest score	6
7	Student_15	80	7th highest score	7
8	Student_6	77	8th highest score	8
9	Student_10	74	9th highest score	9
10	Student_2	72	10th highest score	10

Step	Student	Score	Rank calculation	Rank
11	Student_12	69	11th highest score	11
12	Student_3	68	12th highest score	12
13	Student_7	63	13th highest score	13
14	Student_13	59	14th highest score	14
15	Student_1	56	Lowest score	15

5. Percentile Formula

For this table, percentile is calculated from rank using:

$$\text{Percentile} = ((N - \text{Rank}) / (N - 1)) \times 100$$

Here N = 15, so:

$$\text{Percentile} = ((15 - \text{Rank}) / 14) \times 100$$

Excel formula if Rank is in E2: $= (15 - E2) / (15 - 1)$

Format the Excel result as Percentage.

6. Percentile Calculations — Every Student

Rank	Score	Substitution	Result
1	95	$((15 - 1) / 14) \times 100$	100.00%
2	93	$((15 - 2) / 14) \times 100$	92.86%
3	91	$((15 - 3) / 14) \times 100$	85.71%
4	88	$((15 - 4) / 14) \times 100$	78.57%
5	85	$((15 - 5) / 14) \times 100$	71.43%
6	81	$((15 - 6) / 14) \times 100$	64.29%
7	80	$((15 - 7) / 14) \times 100$	57.14%
8	77	$((15 - 8) / 14) \times 100$	50.00%
9	74	$((15 - 9) / 14) \times 100$	42.86%
10	72	$((15 - 10) / 14) \times 100$	35.71%
11	69	$((15 - 11) / 14) \times 100$	28.57%
12	68	$((15 - 12) / 14) \times 100$	21.43%
13	63	$((15 - 13) / 14) \times 100$	14.29%
14	59	$((15 - 14) / 14) \times 100$	7.14%
15	56	$((15 - 15) / 14) \times 100$	0.00%

7. Detailed Examples

Student_8 — Score 95 — Rank 1

$N = 15$

$\text{Rank} = 1$

$\text{Percentile} = ((15 - 1) / (15 - 1)) \times 100$

$= (14 / 14) \times 100$

$= 1 \times 100$

$= 100.00\%$

Student_14 — Score 93 — Rank 2

$N = 15$

$\text{Rank} = 2$

$\text{Percentile} = ((15 - 2) / (15 - 1)) \times 100$

$= (13 / 14) \times 100$

$= 0.928571 \times 100$

$= 92.86\%$

Student_4 — Score 91 — Rank 3

$N = 15$

$\text{Rank} = 3$

$\text{Percentile} = ((15 - 3) / (15 - 1)) \times 100$

$= (12 / 14) \times 100$

$= 0.857143 \times 100$

$= 85.71\%$

Student_1 — Score 56 — Rank 15

$N = 15$

$\text{Rank} = 15$

$\text{Percentile} = ((15 - 15) / (15 - 1)) \times 100$

$= (0 / 14) \times 100$

$= 0 \times 100$

$= 0.00\%$

8. Final Result Table

Student	Score	Rank	Percentile
Student_1	56	1	100.00%
Student_2	72	6	64.29%
Student_3	68	4	78.57%

Student	Score	Rank	Percentile
Student_4	91	13	14.29%
Student_5	85	11	28.57%
Student_6	77	8	50.00%
Student_7	63	3	85.71%
Student_8	95	15	0.00%
Student_9	81	10	35.71%
Student_10	74	7	57.14%
Student_11	88	12	21.43%
Student_12	69	5	71.43%
Student_13	59	2	92.86%
Student_14	93	14	7.14%
Student_15	80	9	42.86%

Important: This document uses the rank-based percentile formula shown above. It is not the same as Excel's PERCENTILE.INC or PERCENTILE.EXC functions.